

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

POSPrinter (TM-J2000/ TM-J2100)

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Section 1. Introduction

This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-J2000, TM-J2100 Series POS Printers are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

Serial	Parallel	USB	Ethernet
TM-J2000	TM-J2000P	TM-J2000U	TM-J2000E
TM-J2000M	TM-J2000PM	TM-J2000MU	TM-J2000ME
TM-J2100	TM-J2100P	TM-J2100U	TM-J2100E
TM-J2100M	TM-J2100PM	TM-J2100MU	TM-J2100ME

Before reading the manual, see the following explanation about the characteristic of the TM-J2000, TM-J2100 models.

- Station: Receipt (Serial Ink Jet 180 dpi X 180 dpi)
- 2-Color Printing (TM-J2100)

Throughout the manual, the various model names will be referred to as TM-J2000, TM-J2100.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the TM-J2000, TM-J2100 printers.

2.1. References of Firmware Versions

Refer to the release notes (Relnote.txt).

2.2. Settings of DIP Switches

Confirm that the following settings have been made correctly.

1) Serial port

DIP-SW 1

No.	Setting	
1	OFF	Settable
2	ON	Settable
3	OFF	Fixed at OFF
4	OFF	Fixed at OFF
5	OFF	Settable
6	OFF	Settable
7	ON	Settable
8	OFF	Settable

DIP-SW 2

No.	Setting	
1	OFF	Fixed at OFF

Set DIP-SW1-1 (Power button function).

If this DIP Switch is ON, the power button is disabled.

Set DIP-SW1-2 (Interface condition selection).

If this DIP Switch is ON, communications settings are set via DIP Switches. If the DIP Switch is OFF, Memory Switches are used.

Set DIP-SW1-3 (Handshake) to DTR/DSR.

Set DIP-SW1-4 (Bit length) to 8 bits.

Set DIP-SW1-5 to DIP-SW1-8 accordance with the port information.

The described set values are the default values. For the details, refer to the product manual of the POSPrinter. Also, if these settings are changed, make sure to change the port information using the SetupPOS utility.

Do not change the current settings of DIP-SW2-1.

2) Parallel Port

DIP-SW1

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	OFF
8	OFF

Settable

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

DIP-SW2

No.	Setting
1	OFF

Fixed at OFF

Set DIP-SW1-1 (Power button function).

If this DIP Switch is ON, the power button is disabled.

Do not change the current settings of DIP-SW2-1

Make other settings in accordance with the settings described above.

3) USB Port

DIP-SW1

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	OFF
8	OFF

Settable

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

Fixed at OFF

DIP-SW2

No.	Setting
1	OFF

Fixed at OFF

Set DIP-SW1-1 (Power button function).

If this DIP Switch is ON, the power button is disabled.

Do not change the current settings of DIP-SW2-1.

Make other settings in accordance with the settings described above.

4) Ethernet Port

DIP-SW1

No.	Setting	
1	OFF	Settable
2	OFF	Fixed at OFF
3	OFF	Fixed at OFF
4	OFF	Fixed at OFF
5	OFF	Fixed at OFF
6	OFF	Fixed at OFF
7	OFF	Fixed at OFF
8	OFF	Fixed at OFF

DIP-SW2

No.	Setting	
1	OFF	Fixed at OFF

Set DIP-SW1-1 (Power button function).

If this DIP Switch is ON, the power button is disabled.

Do not change the current settings of DIP-SW2-1.

Make other settings in accordance with the settings described above.

2.3. Port Information**1) Port information when serial port is used**

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Baud rate [bps]	2400 ^{*1} , 4800 ^{*1} , 9600, 19200, 38400, 57600 ^{*1} , 115200
Bit length [bit]	8
Parity	NONE, ODD, EVEN
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

^{*1} These Baud rate require setting via Memory Switch.

The default settings are as shown in the following table.

Item	Setting range
Baud rate [bps]	19200
Bit length [bit]	8
Parity	NONE
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	1024
Output interval time [ms]	2500

2) Port information when using parallel port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	1024
Output interval time [ms]	2500

3) Port information when using USB port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output interval time [ms]	0 to 9999

The default setting is as shown in the following table.

Item	Setting range
Output interval time [ms]	2500

4) Port information when using Ethernet port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	1024
Output interval time [ms]	2500

2.4. Device Settings

The following explanation is about the settings for TM-J2000, TM-J2100.

2.4.1. Usable Device Specific Settings

For the TM-J2000, TM-J2100, the following device specific settings are settable by the SetupPOS utility. For the detail, please refer to the Section 2 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”.

Tab	Settings
General	Disable panel buttons
	Assume print complete when data output finishes
	Ignore firmware version check
	Homogenize Error Codes
	Output complete timeout
Paper	Paper Type
	Paper Width [mm]: LineWidth [dot]: LineCharsList
Bitmap	TMFlogo...
	NVRAM
Color Bitmap	Method
	Brightness
	Primary
	Secondary ^{*1}
Status Log	ERROR
	OFFLINE
	Log file name (include full path)
	Maximum file size [KB]
Default Value ^{*2}	Multilingual font

^{*1} Available only for the TM-J2100.

^{*2} Available only for the Multilingual character model.

2.4.2. Paper Type Setting

The TM-J2000, TM-J2100 models support the following paper type. After adding the TM-J2000, TM-J2100 by the SetupPOS utility, open the “Device Specific Settings” dialog box. In the “Paper” tab the paper type could be selected.

The settable paper types are as follows:

- Normal

The default paper type is set to Normal.

2.4.3. Paper Width Setting

The TM-J2000, TM-J2100 models support the following paper width. After adding the TM-J2000, TM-J2100 by the SetupPOS utility, open the “Device Specific Settings” dialog box. In the “Paper” tab the paper width could be selected.

The settable paper widths are as follows:

- 76.0 mm [480 dot]
- 82.5 mm [512 dot]
- 69.5 mm [432 dot]
- 57.5 mm [360 dot]

The default paper width is set to 76.0 mm.

Section 3. Function Details

This section describes the functions of the TM-J2000, TM-J2100 printers in details. Supplementary explanation of the parts not described in detail in the "UPOS" is also given here.

3.1. Property Set Values and Default Values

The following explanation is about the property set values and the default values.

3.1.1. Capability Set Values

The following values are the Capability set values.

Capability Name	Set Value
CapTransaction	TRUE
CapCoverSensor	TRUE
CapConcurrentRecSlp	FALSE
CapConcurrentJrnSlp	FALSE
CapConcurrentJrnRec	FALSE
CapConcurrentPageMode	FALSE
CapCharacterSet	PTR_CCS_ASCII *1
CapMapCharacterSet	FALSE
CapJrnUnderline	FALSE
CapJrnNearEndSensor	FALSE
CapJrnItalic	FALSE
CapJrnEmptySensor	FALSE
CapJrnDwideDhigh	FALSE
CapJrnDwide	FALSE
CapJrnDhigh	FALSE
CapJrnColor	0
CapJrnCartridgeSensor	0
CapJrnBold	FALSE
CapJrn2Color	FALSE
CapJrnPresent	FALSE
CapRecPageMode	TRUE
CapRecUnderline	TRUE
CapRecStamp	FALSE
CapRecRotate180	TRUE

CapRecRight90	TRUE
CapRecPapercut	TRUE
CapRecNearEndSensor	TRUE
CapRecMarkFeed	0
CapRecLeft90	TRUE
CapRecItalic	FALSE
CapRecEmptySensor	TRUE
CapRecDwideDhigh	TRUE
CapRecDwide	TRUE
CapRecDhigh	TRUE
CapRecColor	PTR_COLOR_PRIMARY PTR_COLOR_CUSTOM1 *2
CapRecCartridgeSensor	PTR_CART_REMOVED PTR_CART_EMPTY PTR_CART_NEAREND PTR_CART_CLEANING
CapRecBold	TRUE
CapRecBitmap	TRUE
CapRecBarCode	TRUE
CapRec2Color	TRUE*2
CapRecPresent	TRUE
CapSlpUnderline	FALSE
CapSlpRotate180	FALSE
CapSlpRight90	FALSE
CapSlpNearEndSensor	FALSE
CapSlpLeft90	FALSE
CapSlpItalic	FALSE
CapSlpEmptySensor	FALSE
CapSlpDwideDhigh	FALSE
CapSlpDwide	FALSE
CapSlpDhigh	FALSE
CapSlpColor	0
CapSlpCartridgeSensor	0
CapSlpBothSidesPrint	FALSE
CapSlpBold	FALSE
CapSlpBitmap	FALSE
CapSlpBarCode	FALSE
CapSlp2Color	FALSE
CapSlpFullslip	FALSE
CapSlpPresent	FALSE
CapSlpPageMode	FALSE

*1 If Multilingual character model, "PTR_CCS_KANJI" is set.

*2 Available only for the TM-J2100.

3.1.2. List Properties

The List Properties are explained in the following.

List Property	Settings
CharacterSetList	"255,437,850,852,858,860,863,865,866,998,999,1252" *1
JrnLineCharsList	""
RecLineCharsList (When 82.5 mm is set)	"42,56"
RecLineCharsList (When 76 mm is set)	"40,53"
RecLineCharsList (When 70 mm is set)	"36,48"
RecLineCharsList (When 58 mm is set)	"30,40"
SlpLineCharsList	""
RecBarCodeRotationList	"0,R90, L90, 180"
RecBitmapRotationList	"0,R90, L90, 180"
SlpBarCodeRotationList	""
SlpBitmapRotationList	""
FontTypefaceList	""

*1 If Multilingual character model, "936" or "950" is added to the list.

3.1.3. Width and Height Properties

The width and height properties are explained in the following.

Property	Settings		
	Default Value	Maximum value [dot]	Minimum value [dot]
RecLineSpacing	30	127	0
JrnLineSpacing	X	X	X
SlpLineSpacing	X	X	X
SlpLineHeight [dot]	X		
RecLineHeight [dot]	24,17		
JrnLineHeight [dot]	X		
SlpLineWidth [dot]	X		
RecLineWidth [dot] (When 82.5 mm is set)	512		
RecLineWidth [dot] (When 76 mm is set)	480		
RecLineWidth [dot] (When 70 mm is set)	432		
RecLineWidth [dot] (When 58 mm is set)	360		
JrnLineWidth [dot]	X		
RecSidewaysMaxLines (When 82.5 mm is set)	17		
RecSidewaysMaxLines (When 76 mm is set)	16		
RecSidewaysMaxLines (When 70 mm is set)	14		
RecSidewaysMaxLines (When 58mm is set)	12		
RecSidewaysMaxChars (When 82.5 mm is set / Font A)	(TM-J2000) 69 (TM-J2100) 53		
RecSidewaysMaxChars (When 82.5 mm is set / Font B)	(TM-J2000) 92 (TM-J2100) 71		
RecSidewaysMaxChars (When 76 mm is set / Font A)	(TM-J2000) 74 (TM-J2100) 56		
RecSidewaysMaxChars (When 76 mm is set / Font B)	(TM-J2000) 98 (TM-J2100) 75		
RecSidewaysMaxChars (When 70 mm is set / Font A)	(TM-J2000) 82 (TM-J2100) 63		
RecSidewaysMaxChars (When 70 mm is set / Font B)	(TM-J2000) 109 (TM-J2100) 84		
RecSidewaysMaxChars	(TM-J2000) 98		

(When 58 mm is set / Font A)	(TM-J2100) 76
RecSidewaysMaxChars	(TM-J2000) 131
(When 58 mm is set / Font B)	(TM-J2100) 101
RecLinesToPaperCut	7 ^{*1}
SlpSidewaysMaxLines	X
SlpSidewaysMaxChars	X
SlpMaxLines	X

X : No settings

^{*1} It can be changed by the settings of the RecLineSpacing or the character height.

When the RecLineSpacing is "0", the RecLinesToPaperCut is "-1".

3.1.4. Common Property Strings

The Device information properties are described below.

I/F	DeviceName	DeviceDescription
S	TM-J2000	EPSON TM-J2000 POS Printer
	TM-J2000M	EPSON TM-J2000M POS Printer
	TM-J2100	EPSON TM-J2100 POS Printer
	TM-J2100M	EPSON TM-J2100M POS Printer
P	TM-J2000P	EPSON TM-J2000P POS Printer
	TM-J2000PM	EPSON TM-J2000PM POS Printer
	TM-J2100P	EPSON TM-J2100P POS Printer
	TM-J2100PM	EPSON TM-J2100PM POS Printer
U	TM-J2000U	EPSON TM-J2000U POS Printer
	TM-J2000MU	EPSON TM-J2000MU POS Printer
	TM-J2100U	EPSON TM-J2100U POS Printer
	TM-J2100MU	EPSON TM-J2100MU POS Printer
E	TM-J2000E	EPSON TM-J2000E POS Printer
	TM-J2000ME	EPSON TM-J2000ME POS Printer
	TM-J2100E	EPSON TM-J2100E POS Printer
	TM-J2100ME	EPSON TM-J2100ME POS Printer

I/F indicate the connected interface.

The following is the list of the four connecting interfaces.

S: Serial

P: Parallel

U: USB

E: Ethernet

3.1.5. PageMode Print Properties

The Device information properties are described below.

● TM-J2000

Property	Station ^{*2}		
	Journal	Receipt	Slip
PageModeArea	-	(82.5mm) "512", "832" (76.0mm) "480", "888" (69.5mm) "432", "984" (57.5mm) "360", "1184"	-
PageModeDescriptor ^{*1}	-	BM/BC/BMR/BCR	-

● TM-J2100

Property	Station ^{*2}		
	Journal	Receipt	Slip
PageModeArea	-	(82.5mm) "512", "640" (76.0mm) "480", "680" (69.5mm) "432", "760" (57.5mm) "360", "912"	-
PageModeDescriptor ^{*1}	-	BM/BC/BMR/BCR	-

^{*1} Following setting values are used for the PageModeDescriptor property.

BM : Bitmap printing is available.

BC : Barcode printing is available.

BMR : Rotated printing of bitmap is available.

BCR : Rotated printing of barcode is available.

^{*2} If the Station's CapRecPageMode and/or CapSlpPageMode property values are FALSE, the PageModeArea property shall have " " and the PageModeDescriptor property shall have "0" respectively as a setting value.

3.2. Methods

The following explanation is about supported/unsupported Methods, and the detailed information.

Method	Supported/Unsupported	Compatibility with the PageMode printing
PrintNormal	O	O
PrintTwoNormal	X	X
PrintImmediate	O	O ^{*1}
PrintBarCode	O	O ^{*2}
PrintBitmap	O	O ^{*3}
PrintMemoryBitmap	O	O ^{*3}
CutPaper	O (1~100: One point remains uncut/ Full cut)	X
MarkFeed	X	X
ChangePrintSide	X	X
ValidateData	O	O
TransactionPrint	O	O
SetLogo	O	O
SetBitmap	O	O
RotatePrint	O	X
EndRemoval	X	X
BeginRemoval	X	X
EndInsertion	X	X
BeginInsertion	X	X
ClearPrintArea	O	O
PageModePrint	O	O

O : Supported

X : Unsupported

^{*1} If the specified Station is ready to print, the printing data shall not be stored in the PageMode printing buffer but, instead, go straight to printing. If the Station is not ready to print, an error is returned.

^{*2} If other than "LEFT" is specified for the printing position of barcode, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

^{*3} If other than "LEFT" is specified for the printing position of bitmap, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

3.3. Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

Escape Sequence	TM-J2000	TM-J2100	Compatibility with the PageMode printing
#P	0~100	0~100	X
#fP	0~100	0~100	X
#sP	X	X	X
sL	X	X	X
#B	O	O	O
tL	O	O	O
bL	O	O	O
#R	O	O	O
#lF	0~9999	0~9999	O
#uF	0~ approx. 50 cm	0~ approx. 50 cm	O
#rF Maximum [inch]	X	X	X
#E	0~65535	0~65535	X
#fT	X	X	X
bC	O	O	O
!bC	O	O	O
#uC	1~2	1~2	O
iC	X	X	X
!iC	X	X	X
#rC	1	1~2	O
rvC	O	O	O
!rvC	O	O	O
#sC	X	X	X
#fC	X	X	X
tbC	X	X	X
!tbC	X	X	X
tpC	X	X	X
!tpC	X	X	X
1C	O	O	O
2C	O	O	O
3C	O	O	O
4C	O	O	O
#hC	1~8	1~8	O
#vC	1~8	1~8	O
cA	O	O	O ^{*1}
rA	O	O	O ^{*1}

IA	X	X	X
N	O	O	O

O: Supported

X : Unsupported

Numbers: Settable range

^{*1} Regardless of the PageModeHorizontalPosition property setting, center or right adjust what is to be printed based on the PageModePrintArea property setting in the horizontal direction.

3.4. Printable Barcode Type

The TM-J2000, TM-J2100 models allow the following barcode types.

- Code 128
- Code 128 Parsed
- Code 93
- Codabar
- ITF
- Code 39
- JAN 13 (EAN 13)
- JAN 8 (EAN 8)
- UPC-E
- UPC-A

3.5. Power Condition Reports

The TM-J2000, TM-J2100 models support Power Condition Reports as follows.

Powered on reporting: Supported.

Powered off reporting: Supported.

3.6. Synchronous Processing

The TM-J2000, TM-J2100 models support the Process ID for the Synchronous Processing.

3.7. Printing Positions

The TM-J2000, TM-J2100 models support the function for setting printing position.

Function	Receipt
Left margin	O
Printing Position	O

O: Supported

X: Unsupported

When the left margin setting function is supported, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When the printing position settings are supported, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

3.8. Electronic Logo Function (NVRAM)

TM-J2000 and TM-J2100 models feature an electronic logo function (NVRAM).

There are two ways of using NVRAM explained as follows:

1. Using TMFlogo Utility

Start up TMFlogo utility from “Device Specific Settings” dialog box of SetupPOS utility, and register image files (BMP style) with NVRAM in advance. For the details of the registration, please refer to the “Help” of “TMFlogo utility” and/or “EPSON OPOS ADK MANUAL User’s Manual TMFlogo Utility”.

To print image files registered with NVRAM, please use the either of the following DirectIO:

PTR_DI_FLASH_BITMAP
PTR_DI_FLASH_BITMAP2.

Please refer to the Section 4 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)” for detail.

2. Using SetBitmap Method

Checking “NVRAM” check box on “Bitmap” tab in “Device Specific Settings” dialog box of SetupPOS utility enable to register image files with NVRAM using SetBitmap method.

Regarding the details of image files registration with NVRAM using SetBitmap method, please refer to “Appendix - B: Using NVRAM with SetBitmap method” of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE

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An available NVRAM size for TM-J2000 and TM-J2100 models are 393216 bytes.

A Printer is able to register up to 9 NV Bitmaps. If 9 NV Bitmaps are registered already with the printer, this function is not used with execution of SetBitmap method. If you wish to add new bitmaps, delete registered NV Bitmaps first, then re-execute SetBitmap method.

3.9. Printable Bitmap Types and Sizes

The TM-J2000, TM-J2100 models support the following bitmap commands. For the detail, please refer to the Section 3 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”. The allowance ranges for bitmaps are as follows.

● TM-J2000

Bitmap command type	Allowance range		
	x	y	xy
Download bitmap	1~2040	1~2040	<= 196608
Column Format bitmap	1~4256	1~128	
NV bitmap	1~8192	1~2304	

● TM-J2100

Bitmap command type	Allowance range		
	x	y	xy
Download bitmap	1~24576	1~24576	<= 196480
Column Format Color bitmap	1~4256	1~128	
NV bitmap	1~8192	1~2304	

- Even if meet with the limitation described above, a bitmap that extend the paper width cannot be printed.
- When a height of the raster bitmap expands the value described above, the SO (Service Object) will automatically separate the bitmap data into multiple bitmaps, then print the multiple bitmaps data as one connected bitmap.

3.10. Color Bitmap Printing

The TM-J2000, TM-J2100 models support color bitmap printing.

If a 16-color or a 256-color bitmap file is supplied, the image may be printed using 2 colors. If the original file is a 2-color bitmap, it will be printed in monochrome.

The halftone method may be set using the “Device Specific Settings” dialog box.

3.11. Maintenance Counter

The TM-J2000, TM-J2100 models feature a maintenance counter function for retaining an operation log of the printer. The following chart shows the available maintenance counters for the TM-J2000, TM-J2100.

Counter number Hexadecimal	Counter	Unit	Max. Value	Counter Type
1E	Number of line feeds	Lines	143,165,576	Resettable
1F	Average number of head shots (color 1)	Shots	4,294,967,295	Resettable
20 ^{*1}	Average number of head shots (color 2)	Shots	4,294,967,295	Resettable
21	Number of carriage operations	Times	4,294,967,295	Resettable
22	Number of pump operations	Times	4,294,967,295	Resettable
32	Number of auto-cutter operations	Times	4,294,967,295	Resettable
46	Uptime of product	Hours	71,582,788	Resettable
9E	Number of line feeds	Lines	143,165,576	Cumulative
9F	Average number of head shots (color 1)	Shots	4,294,967,295	Cumulative
A0 ^{*1}	Average number of head shots (color 2)	Shots	4,294,967,295	Cumulative
A1	Number of carriage operations	Times	4,294,967,295	Cumulative
A2	Number of pump operations	Times	4,294,967,295	Cumulative

B2	Number of auto-cutter operations	Times	4,294,967,295	Cumulative
C6	Uptime of product	Hours	71,582,788	Cumulative

^{*1} Available only for the TM-J2100.

3.12. Automatic Recovery Function

The TM-J2000, TM-J2100 models feature a function for automatic recovery when the power is turned on again after an interruption of power. Recovery processing is performed automatically when the printer's power is turned on again after an interruption. The recovery processing restores the printer to the condition it was in before the power was turned off.

3.13. Output without Flow Control on the USB/Ethernet Interfaces

The TM-J2000, TM-J2100 models support outputting without flow control on the USB/Ethernet interfaces.

Section 4. Warnings

This section describes precautions in use of TM-J2000, TM-J2100.

- When parallel I/F is used with Windows 2000, Windows XP or Windows Vista, please set Busy Condition of Memory SW1-3 to ON (Buffer full).